Better Forecasts, Higher Trading Profits

Case study: Battery Energy Storage System Trading in Germany

Summary

Battery Energy Storage Systems (“BESS”) are booming around Europe, based on the high expected arbitrage margins achievable by trading BESS assets in different short-term energy and capacity markets.

BESS traders achieve trading margins in several different markets, including the Day-Ahead Auction (DAA), intraday markets, and balancing markets. This paper focuses on the margin achievable in the DAA. The DAA is the largest and most liquid short term power market. Currently in Germany, BESS traders typically count on the DAA for around 20-30% of their annual BESS trading margin. So the DAA is a logical place to start for an analysis of forecast trading value.

As in all markets, in the DAA, accurate forecasting plays an important part in the success of BESS trading. With better forecasts, bids and offers in the DAA markets can be improved in terms of timing, quantity and price.

Yet many BESS optimizers lack a transparent, quantified perspective on *how much* the accuracy of their price forecasts affects their BESS trading results.

To increase transparency, the FlexUp team built a tool to simulate and quantify the incremental trading margin from trading a 1MW / 2MWh BESS in the German DAA, using different forecasts. We compare the trades and resulting margin based on DAA price forecasts from a commercial price forecaster, Electricity Maps, (“E-Maps”) with the trades and resulting margin from a “naive” daily Persistence Forecast (24-hour delayed signal: “Today is like yesterday”). We also compare E-Maps-based trading results to the resulting margin from Perfect Foresight (= actual prices).

Better forecasts enable significantly better trading results. Trading based on the forecasts from E-Maps improves DAA margin by over **€5000 / MW / yr** vs. trading based on the Persistence Forecast. For a 100MW / 200MWh BESS, this forecast improvement would then yield **€500 000** per year additional margins, just from DAA market trading.

Nevertheless, there is still significant additional value for forecasters and traders in further improving the DAA price forecasting. Compared with Perfect Foresight, the Electricity Maps Forecast still leaves over 5500 EUR / MW / year of additional upside.

Methodology

To assess the value of forecasting in BESS trading, our tool runs the same time-arbitrage trading strategy in the DAA with three different forecasts and compares the resulting annual trading margin.

We compare the following three forecasts:

1) **Persistence Forecast**: Vector of prices for the 96 periods in day D = Vector of Vector of prices for the 96 periods in day D-1.
2) **E-Maps Forecast**: Day ahead forecast published at D-1 11AM, i.e. 1 hour before the DAA gate closure.
3) **Perfect Foresight**: The forecast equals the actual market results, so that the forecasting error equals zero.

Based on this forecast, we simulated a simple time-arbitrage strategy in the Day-Ahead-Auction in the DE-LU zone (basically: buy/charge during low-price periods and sell/discharge during high-price periods, pocket the difference). The strategy is simulated with a single-step linear optimization, which sets a charge/discharge (i.e. buy/sell) schedule, given expected prices, to maximize trading margin.

To account for the effects of forecast error, the optimizer computes the trading decisions using the forecasted prices, while the resulting trading margins are calculated using actual prices.

To compare the impact of forecasting errors, we run the identical daily trading algorithm three times—once with Persistence Forecast, once with E-Maps, once with Perfect Foresight. Then we sum up the days and compare the annual trading margin delta per MW.

Trading simulation inputs

The table below lists the assumptions used for each of the trading scenarios. In summary, each scenario uses the same technical constraints, while varying the DAA price expectations.

		Persistence Forecast	**Electricity Maps Forecast**	**Perfect Foresight**
1	Power capacity	1.0 MW		
2	Energy capacity	2.0 MWh		
3	SOC range	0%-100%		
4	Max. cycles per day	2.0		

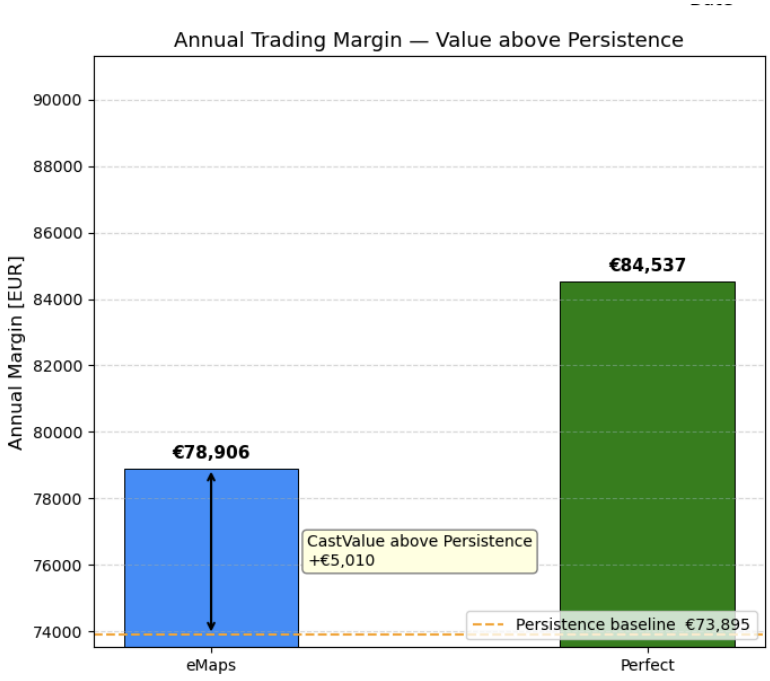
5	Starting / Ending SOC	0 / 0 (each day starts and ends with SOC = 0)		
6	DAA price forecasts used for charge/discharge decisions	Persistence Forecast	Electricity Maps Forecast D-1, 11am	Actual prices

Physical effects such as cycle losses, SOC-measurement errors, line losses are not modelled.

Annual results

The graph below shows the resulting difference to the Persistence Baseline in annual trading margin from the DAA market only, depending on the price forecasts used.

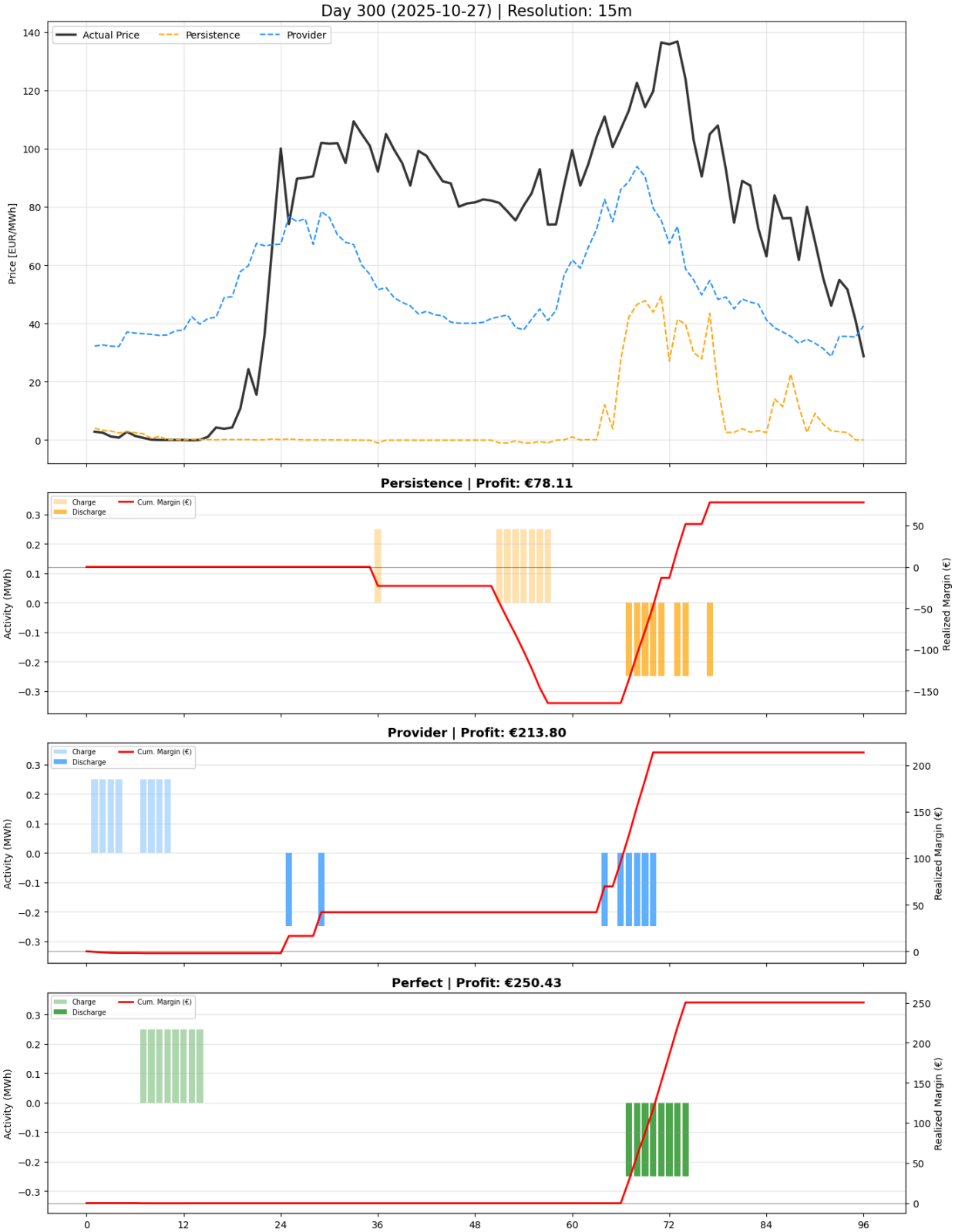
Even excluding all other markets, a portfolio of 100MW/200MWh of batteries could generate over €500 000 per year more if based on forecasts of Electricity Maps than on the benchmark Persistence Forecast.



At the same time, trades based on Perfect Foresight would still generate € 550 000 additionally for the 100MW/200MWh system. Thus significant further margin potential from better forecast accuracy remains.

Example Day Evaluation

A deep dive in an example day shows clearly how inaccurate price forecasts lead to suboptimal charge/discharge decisions and lower daily arbitrage margins for the BESS trader.


Day 300 (2025-10-27) | Resolution: 15m
Actual Price
Persistence
Provider
Price [EUR/MWh]
Persistence | Profit: €78.11
Provider | Profit: €213.80
Perfect | Profit: €250.43
Charge
Discharge
Cum. Margin (€)
Activity (MWh)
Realized Margin (€)

A detailed evaluation of the trading decisions of each of the three “traders” shows how imperfect forecasts lead to suboptimal decisions and lost trading margin:

1. **“Persistence” Trader**: The trader working with the Persistence Forecast charges its BESS much too late in the day, when actual prices are already much higher than forecast. Even though Persistence sells/discharges at roughly the right time and price, the suboptimally high charging cost means this trader can no longer make up for his/her poor charging decisions. His/her daily profit is only €78.11/MW, capturing only 31% of available arbitrage value.
2. **“Provider Forecast” Trader**: The Trader working with the Electricity Maps Forecast buys a little too early, and does not perfectly capture zero prices, because the Provider’s Forecast shows higher and increasing prices in the night and early morning, giving the Trader a false incentive to buy too early. Still this is only a small setback and the Trader continues the day with a charge BESS and very low charging cost. However, because this Trader’s forecast has missed the afternoon price peak, the Trader sells some electricity during the morning peak, thus missing some of the revenue available in the afternoon price peak. Daily profit: €213.80, a respectable, but not perfect 85% of optimal.
3. **“Perfect Foresight” Trader**: The Perfect Foresight Trader has an unfair advantage: He charges at the lowest periods and forecasts correctly that the highest prices are to be achieved during the afternoon peak. Therefore this trader keeps his BESS fully charged throughout the morning price peak, and discharges only during the optimal periods in the late afternoon. Daily profit: €250.43 (100%!).

In short, this example shows that differences in forecast accuracy can have a significant impact on trading margins. Better forecasts are the “jet fuel” for traders/algos which intend to outperform their peers and approach optimal results for themselves and their clients.

Limitations and extensions

To show the basic dynamics of how better forecasting drives BESS trading results, this analysis is purposely quite simple. As such, it is important to point out some of the main limitations of this “v1.0” analysis:

- The model assumes guaranteed execution. In reality, the forecast error can cause the trader to submit out-of-the-money, non-clearing bids and offers, leading to missed fills when unexpectedly no transaction is done for a certain period. For example, when a trader’s forecast prices in low-price (“charging”) periods are below the market clearing price, this could cause a trader to bid below the clearing price and thus not actually do the transaction and charge the BESS. These non-clearing bids/offers lead to missed margin and additional uncertainty from imbalanced positions that need to be dealt with in the intraday markets at unknown prices. These effects are not included in this analysis. If they were, the value of accurate forecasting would likely increase.

- This analysis only covers physical time-arbitrage within the DAA, and not several other markets (such as intraday-auction, intraday continuous or ancillary services), nor other trading opportunities (such as cross-market arbitrage (e.g. DAA vs. IDC), real-option-based strategies ("virtual cycling"), or any other strategies. Many more arbitrage opportunities exist for a BESS that is traded cross-market in Germany. Improved forecasting is key to outperforming peers in all of these markets.

Given these limitations, FlexUp will continue to work on the forecast value tool, specifically focusing on the following extensions / improvements:

- Incorporating the effect of non-clearing bids and resulting "squeezes"
- Covering additional markets, such as intraday and reserve markets
- Replicating the analysis for other geographies, with different general forecasting accuracy levels. Germany tends to have high forecast errors given the renewable share)
- Comparing additional forecasters beyond Electricity Maps to benchmark forecasters against their peers, not just a naive Persistence Forecast
- Replicating the analysis for different trading contexts, including cross-border / connector arbitrage.

These interesting extensions will further generate transparency in price forecasting and power markets.

Further information

This whitepaper is a collaboration between FlexUp GmbH and Electricity Maps ApS.

FlexUp GmbH operates the Teamcast collaborative forecasting platform. This platform collects, compares and ensembles day-ahead price forecasts from many different provieers in order to create transparency in forecasting performance and use the "wisdom of the crowd" to build better day ahead price forecasts. Forecast accuracy is a key factor in keeping electricity markets functioning smoothly, reducing price volatility and making the energy transition affordable in Europe. For further information contact: hello@flexup.pro or visit www.teamcast.flexup.pro .

Electricity Maps has built **t**he world's most comprehensive electricity data platform. Electricity Maps provides global access to electricity mix, prices, load, and carbon intensity. Data is available in real-time, historically and forecasted. Electricity Maps gives you unprecedented insights into the grid, with seamless access to global data at the highest granularity. For further information visit https://www.electricitymaps.com/ or get in touch here: https://www.electricitymaps.com/contact

Further details about this tool and further analysis is available in our Github Repo.